



1. Description

1.1. Project

Project Name	Sp3ctra
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	03/14/2026

1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H747/757
MCU name	STM32H747XIHx
MCU Package	TFBGA240
MCU Pin number	265

1.3. Core(s) information

Core(s)	ARM Cortex-M7 ARM Cortex-M4
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2. Pinout Configuration



TFBGA240 +25 (Top view)

3. Pins Configuration

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSS	Power		
A2	PI6	I/O	DCMI_D6	
A3	PI5	I/O	DCMI_VSYNC	
A4	PI4	I/O	DCMI_D5	
A5	PB5	I/O	SPI1_MOSI	
A6	VDDLDO	Power		
A7	VCAP	Power		
A8	PK5 *	I/O	GPIO_Output	ETH_RST
A9	PG10	I/O	DCMI_D2	
A10	PG9	I/O	SPI1_MISO	
A14	PA15 (JTDI) *	I/O	GPIO_Input	MEMS_FSYNC
A17	VSS	Power		
B1	VBAT	Power		
B2	VSS	Power		
B3	PI7	I/O	DCMI_D7	
B4	PE1	I/O	FMC_NBL1	
B5	PB6	I/O	USART1_TX	
B6	VSS	Power		
B9	PG11	I/O	ETH_TX_EN	
B12	PD3	I/O	SPI2_SCK	
B13	PC11	I/O	DCMI_D4	
B14	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
B15	PI2	I/O	TIM8_CH4	XLSYNC
C3	PE2	I/O	SPI4_SCK	
C4	PE0	I/O	FMC_NBL0	
C5	PB7	I/O	USART1_RX	
C6	PB3 (JTDO/TRACESWO)	I/O	SPI1_SCK	
C10	VSS	Power		
C13	VSS	Power		
C15	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
C16	VSS	Power		
C17	VDDLDO	Power		
D1	PE5	I/O	SPI4_MISO	
D2	PE4	I/O	SPI4_NSS	
D4	PB9	I/O	SPI2_NSS	MEMS_CS
D6	PG15	I/O	FMC_SDNCS	

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
D8	PG14	I/O	ETH_TXD1	
D9	PG13	I/O	ETH_TXD0	
D13	PD0	I/O	FMC_D2	
D14	PA10	I/O	DCMI_D1	
D15	PA9	I/O	TIM1_CH2	CIS_CP
D17	VCAP	Power		
E1	VLXSMPS	Power		
E5	PE6	I/O	SPI4_MOSI	
E6	VDD	Power		
E7	PDR_ON	Power		
E8	BOOT0	Boot		
E9	VDD	Power		
E11	VDD	Power		
E12	PD1	I/O	FMC_D3	
E13	PC8	I/O	TIM8_CH3	CIS_SP
E14	PC9	I/O	DCMI_D3	
E15	PA8	I/O	I2C3_SCL	
E16	PA12 *	I/O	GPIO_Output	CIS_RS
F1	VDDSMPS	Power		
F2	VSSSMPS	Power		
F5	VDD	Power		
F14	PC6	I/O	TIM3_CH1	CIS_LED_B
F15	PG8	I/O	FMC_SDCLK	
F17	VDD33_USB	Power		
G1	PF2	I/O	FMC_A2	
G2	VFBSMPS	Power		
G3	PF1	I/O	FMC_A1	
G4	PF0	I/O	FMC_A0	
G5	VDD	Power		
G7	VSS	Power		
G8	VSS	Power		
G9	VSS	Power		
G10	VSS	Power		
G11	VSS	Power		
G13	VDD	Power		
G14	PG5	I/O	FMC_BA1	
G15	PG6 *	I/O	GPIO_Output	EN_12V
G16	VSS	Power		
G17	VDD50_USB	Power		

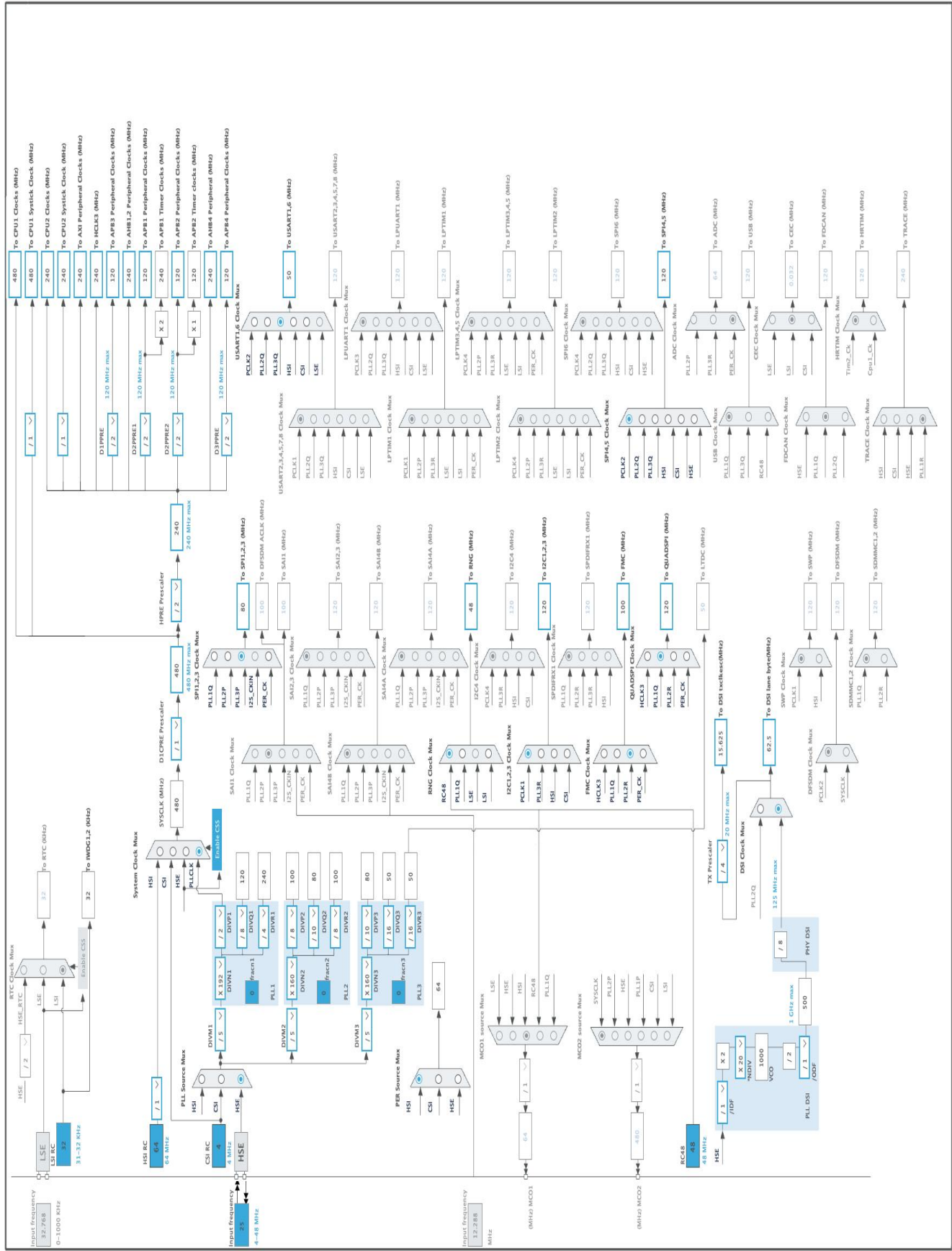
Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
H4	PF3	I/O	FMC_A3	
H5	VDD	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VSS	Power		
H11	VSS	Power		
H13	VDD	Power		
H14	PG4	I/O	FMC_BA0	
H16	PG2	I/O	FMC_A12	
H17	PK2 *	I/O	GPIO_Output	EN_5V
J1	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
J2	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
J3	VSS	Power		
J4	PF5	I/O	FMC_A5	
J5	PF4	I/O	FMC_A4	
J7	VSS	Power		
J8	VSS	Power		
J9	VSS	Power		
J10	VSS	Power		
J11	VSS	Power		
J13	VDD	Power		
J16	VSS	Power		
J17	VSS	Power		
K1	NRST	Reset		
K2	PF6	I/O	QUADSPI_BK1_IO3	
K3	PF7	I/O	QUADSPI_BK1_IO2	
K4	PF8	I/O	QUADSPI_BK1_IO0	
K5	VDD	Power		
K7	VSS	Power		
K8	VSS	Power		
K9	VSS	Power		
K10	VSS	Power		
K11	VSS	Power		
K13	VDD	Power		
K15	VSSDSI	Power		
K16	DSI_D1P	MonolO	DSIHOST_D1P	
K17	DSI_D1N	MonolO	DSIHOST_D1N	
L1	VDDA	Power		

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
L2	PC0	I/O	FMC_SDNWE	
L3	PF10	I/O	QUADSPI_CLK	
L4	PF9	I/O	QUADSPI_BK1_IO1	
L5	VDD	Power		
L7	VSS	Power		
L8	VSS	Power		
L9	VSS	Power		
L10	VSS	Power		
L11	VSS	Power		
L13	VDD	Power		
L15	VSSDSI	Power		
L16	DSI_CKP	MonoIO	DSIHOST_CKP	
L17	DSI_CKN	MonoIO	DSIHOST_CKN	
M2	PC1	I/O	ETH_MDC	
M3	PC2	I/O	FMC_SDNE0	
M4	PC3	I/O	FMC_SDCKE0	
M5	VDD	Power		
M13	VDD	Power		
M15	VSSDSI	Power		
M16	DSI_D0P	MonoIO	DSIHOST_D0P	
M17	DSI_D0N	MonoIO	DSIHOST_D0N	
N1	VREF-	Power		
N3	PA2	I/O	ETH_MDIO	
N4	PA1	I/O	ETH_REF_CLK	
N7	VDD	Power		
N8	VDD	Power		
N9	PE10	I/O	FMC_D7	
N10	VDD	Power		
N11	VDD	Power		
N12	VDD	Power		
N13	PJ8	I/O	GPIO_EXTI8	MEMS_INT
N16	VSS	Power		
N17	VCAPDSI	Power		
P1	VSSA	Power		
P3	PH4	I/O	I2C2_SCL	
P4	PH5	I/O	I2C2_SDA	
P7	PF13	I/O	FMC_A7	
P8	PF14	I/O	FMC_A8	
P9	PE9	I/O	FMC_D6	

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
P10	PE11	I/O	FMC_D8	
P11	PB10	I/O	QUADSPI_BK1_NCS	
P15	PD15	I/O	FMC_D1	
P16	PD14	I/O	FMC_D0	
P17	VDDDSI	Power		
R3	PA6	I/O	DCMI_PIXCLK	
R4	VSS	Power		
R5	PA7	I/O	ETH_CRS_DV	
R7	PF12	I/O	FMC_A6	
R8	VSS	Power		
R9	PF15	I/O	FMC_A9	
R10	PE12	I/O	FMC_D9	
R11	PE15	I/O	FMC_D12	
R13	PH9	I/O	DCMI_D0	
R14	PH12	I/O	TIM5_CH3	CIS_LED_G
R17	PD13	I/O	TIM4_CH2	CIS_LED_R
T4	PC4	I/O	ETH_RXD0	
T7	PF11	I/O	FMC_SDNRAS	
T8	PG0	I/O	FMC_A10	
T9	PE8	I/O	FMC_D5	
T10	PE13	I/O	FMC_D10	
T12	VSS	Power		
T13	PH8	I/O	I2C3_SDA	
T15	PB15	I/O	SPI2_MOSI	
T16	PD10	I/O	FMC_D15	
T17	PD9	I/O	FMC_D14	
U1	VSS	Power		
U3	PA4	I/O	DCMI_HSYNC	
U4	PC5	I/O	ETH_RXD1	
U8	PG1	I/O	FMC_A11	
U9	PE7	I/O	FMC_D4	
U10	PE14	I/O	FMC_D11	
U11	VCAP	Power		
U12	VDDLDO	Power		
U15	PB14	I/O	SPI2_MISO	
U16	PD8	I/O	FMC_D13	
U17	VSS	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H747/757
MCU	STM32H747XIHx
Datasheet	DS12930 Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(DD36000)
Capacity	36000.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	450.0 mA
Max Pulse Current	1000.0 mA
Cells in series	1
Cells in parallel	1

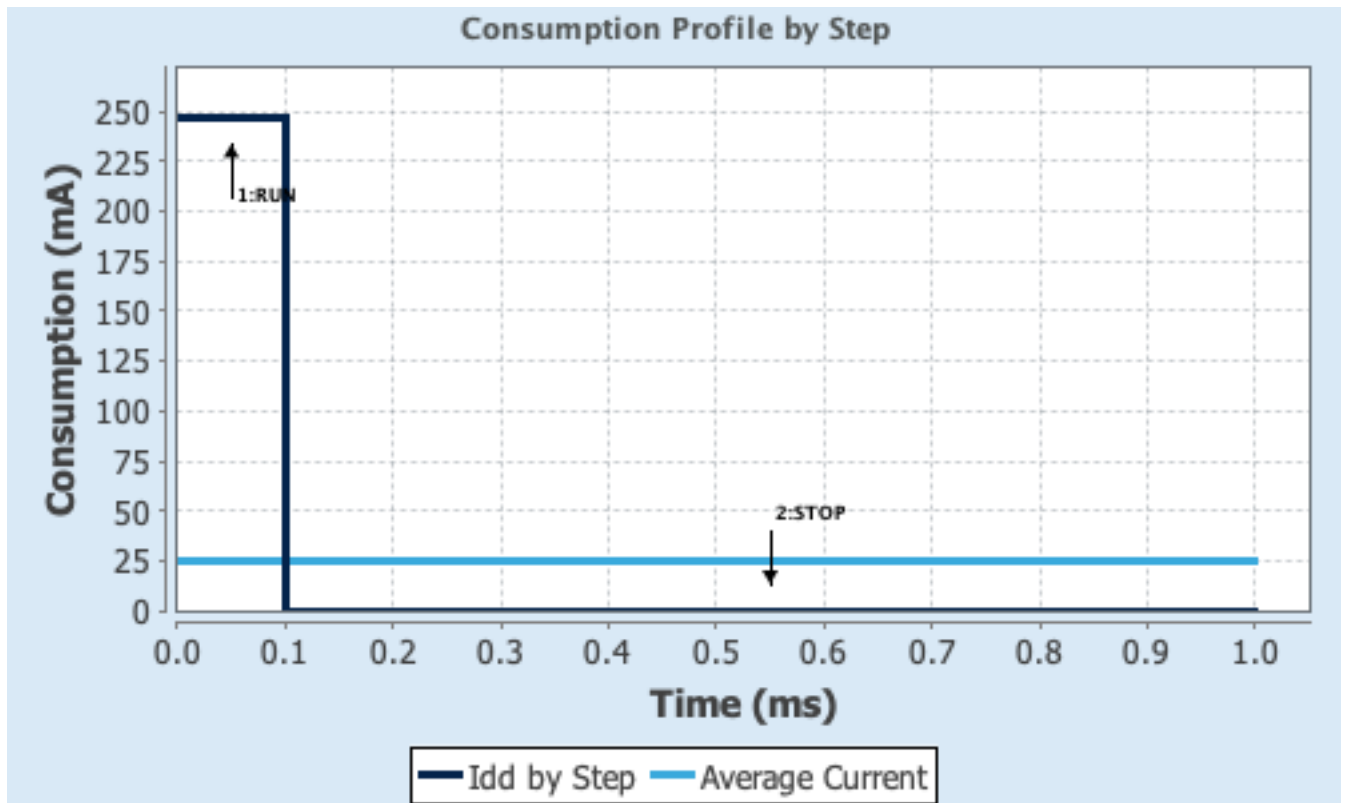
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN/CRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	CM7: ITCM/Cache / CM4: FLASH_B/ART	CM7: NA / CM4: NA
CM7 Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL ALL_IPs_ON	LSE Flash-ON
CM4 Frequency	240 MHz	0 Hz
Clock Source Frequency	25 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	247 mA	145 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	24.83 mA
Battery Life	1 month, 29 days, 21 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	Sp3ctra
Project Folder	/Users/zhonx/Documents/Projets_PRE/CISYNTH/CIS/Electronique/Sp3ctra_CIS
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_H7 V1.12.1
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x400
Minimum Stack Size	0x800

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls ARM Cortex-M7

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_DMA_Init	DMA
3	SystemClock_Config	RCC
4	MX_MDMA_Init	MDMA
5	MX_FMC_Init	FMC
6	MX_USART1_UART_Init	USART1
7	MX_TIM1_Init	TIM1
8	MX_TIM3_Init	TIM3
9	MX_TIM4_Init	TIM4
10	MX_TIM5_Init	TIM5
11	MX_TIM8_Init	TIM8

Rank	Function Name	Peripheral Instance Name
12	MX_RNG_Init	RNG
13	MX_CRC_Init	CRC
14	MX_LWIP_Init	LWIP
15	MX_QUADSPI_Init	QUADSPI
16	MX_SPI2_Init	SPI2
17	MX_FREERTOS_Init	FREERTOS_M7
18	MX_FATFS_Init	FATFS_M7
19	MX_TIM6_Init	TIM6
20	MX_IWDG1_Init	IWDG1
21	MX_DMA2D_Init	DMA2D
22	MX_DCMI_Init	DCMI
23	MX_I2C2_Init	I2C2
24	MX_SPI1_Init	SPI1
25	MX_SPI4_Init	SPI4

2.4. Advanced Settings - Generated Function Calls ARM Cortex-M4

Rank	Function Name	Peripheral Instance Name
1	MX_DMA_Init	DMA
2	MX_GPIO_Init	GPIO
3	MX_MDMA_Init	MDMA
4	MX_FMC_Init	FMC
5	MX_USART1_UART_Init	USART1
6	MX_TIM12_Init	TIM12
7	MX_DSIHOST_DSI_Init	DSIHOST
8	MX_I2C3_Init	I2C3

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7

3.1.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Speculation default mode Settings:

Speculation default mode	Disabled
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Cortex Interface Settings:

CPU ICache	Enabled *
CPU DCache	Enabled *

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers *
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Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x0 *
MPU Region Size	4GB *
MPU SubRegion Disable	0x87 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE *
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x30020000 *
MPU Region Size	128KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE

MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x30040000 *
MPU Region Size	512B *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x24000000 *
MPU Region Size	4KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	ENABLE *
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x24000000 *
MPU Region Size	2KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 6 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

3.2. CRC

mode: Activated

3.2.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M7

Initialized Context: Cortex-M7

Power Domain: D3

Basic Parameters:

Default Polynomial State Enable

Default Init Value State Enable

Advanced Parameters:

Input Data Inversion Mode None

Output Data Inversion Mode Disable

Input Data Format **Words ***

3.3. DCMI

DCMI: Slave 8 bits External Synchro

3.3.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Mode Config:

Pixel clock polarity	Active on Rising edge *
Vertical synchronization polarity	Active High *
Horizontal synchronization polarity	Active High *
Frequency of frame capture	All frames are captured
JPEG mode	Disabled

Interface Capture Config:

Byte Select Mode	Interface captures all received bytes
Line Select Mode	Interface captures all received lines

3.4. DEBUG

Debug: Serial Wire

3.4.1. Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	

3.5. DMA2D

mode: Activated

3.5.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M7
 Initialized Context: Cortex-M7
 Power Domain: D1

Basic Parameters:

Transfer Mode Memory to Memory
 Color Mode ARGB8888
 Output Offset 0

Foreground layer Configuration:

DMA2D Input Color Mode ARGB8888
 DMA2D ALPHA MODE No modification of the alpha channel value
 Input Alpha 0
 Input Offset 0
 DMA2D ALPHA Inversion Regular Alpha
 DMA2D Red and Blue swap Regular mode (RGB or ARGB)
 DMA2D Chroma Sub-Sampling Mode No chroma sub-sampling 4:4:4

3.6. DSIHOST

DSIHost: Command Mode via APB Interface

3.6.1. DSI Clocks:

Core(s) Settings:

Context(s): Cortex-M4
 Initialized Context: Cortex-M4
 Power Domain: D1

from PLLDSI:

PlIndiv 20
 Pllof DSI_PLL_OUT_DIV1
 Plidf DSI_PLL_IN_DIV1
 High Speed Clock - PLLDSI Output 500000
 Lane Byte Clock 62500
 Tx Escape Ckdiv 4
 Transmission Escape Clock 15625
 Time Out Clock 62500

from PLL2Q:

Lane Byte Clock 80000
 Transmission Escape Clock 20000
 Time Out Clock 80000

3.6.2. Timeout Counters:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D1

Time Out Clock Setting:

Time Out Clock Divider	1
Time Out Clock - from PLLDSI	62500

Contention Error Detection:

High-speed transmission time-out	0
No High-speed transmission time-out	
Low-power reception time-out	0
No Low-power reception time-out	

Peripheral Response:

High-speed read time-out	0
Resulting Timing High-speed read time-out	0
Low-power read time-out	0
Resulting Timing Low-power read time-out	0
High-speed write time-out	0
Resulting Timing High-speed write time-out	0
Low-power write time-out	0
Resulting Timing Low-power write time-out	0
BTA time-out	0
Resulting Timing for BTA time-out	0

3.6.3. Data and Clock Lanes:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D1

Basic Settings:

Number of Lanes	One Data Lane
Automatic Clock Lane Control	Clock lane is always provided
Bus Turn Around Request is	Enabled

Flow Control - Configuration:

CRC Reception	Disabled
ECC Reception	Disabled
EoTP Reception is	Disabled
EoTP Transmission is	Disabled
Acknowledge Request after Each Transmission	Disabled

Flow Control - Packet Analyzer Configuration:

CRC Error Interrupt	Disable
ECC Errors Interrupt	Disable
EoTP Error Interrupt	Disable
Packet Size Error Interrupt	Disable
Acknowledge Errors Interrupt	Disable
PHY related Errors Interrupt	Disable

3.6.4. PHY Timings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D1

3.6.5. Commands:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D1

APB Interface Error Configuration:

Generic Command Error Interrupt	Disable
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Transmission Mode for Commands::

Generic Short Write Zero Parameter	High Speed Transmission
Generic Short Write One Parameter	High Speed Transmission
Generic Short Write Two parameters	High Speed Transmission
Generic Short Read Zero parameter	High Speed Transmission
Generic Short Read One parameter	High Speed Transmission
Generic Short Read Two parameters	High Speed Transmission
Generic Long Write	High Speed Transmission
DCS Short Write Zero parameter	High Speed Transmission
DCS Short Write One parameter	High Speed Transmission

DCS Short Read Zero parameter	High Speed Transmission
DCS Long Write	High Speed Transmission
Maximum Read Packet Size Command	High Speed Transmission

3.7. ETH

Mode: RMII

3.7.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

General : Ethernet Configuration:

Warning	Ethernet RX and Tx descriptors needs to be placed in a RAM memory accessible by the dedicated Ethernet DMA
Note	PHY Driver must be configured from the LwIP 'Platform Settings' top right tab
Ethernet MAC Address	00:80:E1:00:00:00
Tx Descriptor Length	4
First Tx Descriptor Address	0x30040100 *
Rx Descriptor Length	4
First Rx Descriptor Address	0x30040000 *
Rx Buffers Address	0x30040200 *
Rx Buffers Length	1536

3.8. FMC

SDRAM 1

Clock and chip enable: SDCKE0+SDNE0

Internal bank number: 4 banks

Address: 13 bits

Data: 16 bits

Byte enable: 16-bit byte enable

3.8.1. SDRAM 1:

Core(s) Settings:

Context(s):	Cortex-M7
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Initialized Context:	Cortex-M4 Cortex-M7
Power Domain:	D1
SDRAM control:	
Bank	SDRAM bank 1
Number of column address bits	9 bits *
Number of row address bits	13 bits
CAS latency	3 memory clock cycles *
Write protection	Disabled
SDRAM common clock	Disabled
SDRAM common burst read	Disabled
SDRAM common read pipe delay	1 HCLK clock cycle *
SDRAM timing in memory clock cycles:	
Load mode register to active delay	2 *
Exit self-refresh delay	7 *
Self-refresh time	5 *
SDRAM common row cycle delay	6 *
Write recovery time	2 *
SDRAM common row precharge delay	2 *
Row to column delay	16

3.8.2. Bank Mapping:

Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	D1

Mapping parameters:

FMC bank mapping	Default mapping
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3.9. I2C2

I2C: I2C

3.9.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M7
Initialized Context: Cortex-M7
Power Domain: D2

Timing configuration:

Custom Timing Disabled
I2C Speed Mode Standard Mode
I2C Speed Frequency (KHz) 100
Rise Time (ns) 0
Fall Time (ns) 0
Coefficient of Digital Filter 0
Analog Filter Enabled
Timing **0x307075B1 ***

Slave Features:

Clock No Stretch Mode Disabled
General Call Address Detection Disabled
Primary Address Length selection 7-bit
Dual Address Acknowledged Disabled
Primary slave address 0

3.10. I2C3

I2C: I2C

3.10.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M4
Initialized Context: Cortex-M4
Power Domain: D2

Timing configuration:

Custom Timing Disabled
I2C Speed Mode Standard Mode
I2C Speed Frequency (KHz) 100
Rise Time (ns) 0
Fall Time (ns) 0
Coefficient of Digital Filter 0
Analog Filter Enabled

Timing **0x307075B1 ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.11. IWDG1

mode: Activated

3.11.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D3

Watchdog Clocking:

IWDG counter clock prescaler	128 *
IWDG window value	0xff
IWDG down-counter reload value	(32000*10000)/(128*1000) *

3.12. QUADSPI

QuadSPI Mode: Bank1 with Quad SPI Lines

3.12.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

General Parameters:

Clock Prescaler	1 *
Fifo Threshold	16 *
Sample Shifting	Sample Shifting Half Cycle *
Flash Size	1

Chip Select High Time	4 Cycles *
Clock Mode	Low
Flash ID	Flash ID 1
Dual Flash	Disabled

3.13. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.13.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	D3

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 4 and 8 MHz
PLL2 input frequency range	Between 4 and 8 MHz
PLL3 input frequency range	Between 4 and 8 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	Wide VCO range

3.14. RNG

mode: Activated

3.14.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2
Clock Error Detection	Enable

3.15. SPI1

Mode: Full-Duplex Master

3.15.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	40.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle

Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.16. SPI2

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.16.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	4 *
Baud Rate	20.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Output Hardware
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	10 Cycle *
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Enable *
Master Keep Io State	Master Keep Io State Enable *
IO Swap	Disabled

3.17. SPI4

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.17.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	60.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Output Hardware
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.18. SYS

Timebase Source: TIM2

3.18.1. Core(s) Settings:

Context(s): Cortex-M7
 Initialized Context: Cortex-M7
 Power Domain:

3.19. SYS_M4

Timebase Source: SysTick

3.19.1. Core(s) Settings:

Context(s): Cortex-M4
 Initialized Context: Cortex-M4
 Power Domain:

3.20. TIM1

Channel1: PWM Generation No Output

Channel2: PWM Generation CH2

3.20.1. Parameter Settings:

Core(s) Settings:

Context(s): Cortex-M7
 Initialized Context: Cortex-M7
 Power Domain: D2

Counter Settings:

Prescaler (PSC - 16 bits value) **prescalerValue ***
 Counter Mode Up
 Counter Period (AutoReload Register - 16 bits value) **counterPeriod ***
 Internal Clock Division (CKD) No Division
 Repetition Counter (RCR - 16 bits value) 0
 auto-reload preload **Enable ***

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit) **Enable (Trigger delayed for master/slaves simultaneous start) ***
 Trigger Event Selection TRGO **Output Compare (OC1REF) ***

Trigger Event Selection TRGO2

Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
--------------------	---------

PWM Generation Channel 1:

Mode	PWM mode 2 *
Pulse (16 bits value)	pulseValueCH1 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	pulseValueCH2 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.21. TIM3

Slave Mode: External Clock Mode 1

Trigger Source: ITR0

Channel1: PWM Generation CH1

3.21.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	prescalerValue *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	counterPeriod *
Internal Clock Division (CKD)	No Division
auto-reload preload	Enable *
Slave Mode Controller	ETR mode 1

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
--------------------	---------

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	pulseValue *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.22. TIM4

Slave Mode: External Clock Mode 1

Trigger Source: ITR0

Channel2: PWM Generation CH2

3.22.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	prescalerValue *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	counterPeriod *
Internal Clock Division (CKD)	No Division
auto-reload preload	Enable *
Slave Mode Controller	ETR mode 1

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
--------------------	---------

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	pulseValue *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.23. TIM5

Slave Mode: External Clock Mode 1

Trigger Source: ITR0

Channel3: PWM Generation CH3

3.23.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	prescalerValue *
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	counterPeriod *
Internal Clock Division (CKD)	No Division
auto-reload preload	Enable *
Slave Mode Controller	ETR mode 1

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
--------------------	---------

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (32 bits value)	pulseValue *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.24. TIM6

mode: Activated

3.24.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	prescalerValue *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	counterPeriod *
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)
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3.25. TIM8

Slave Mode: External Clock Mode 1

Trigger Source: ITR0

Channel3: PWM Generation CH3

Channel4: PWM Generation CH4

3.25.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	prescalerValue *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	counterPeriod *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Enable *
Slave Mode Controller	ETR mode 1

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable

- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
--------------------	---------

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	pulseValue *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.26. TIM12

mode: Clock Source

3.26.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M4
Initialized Context:	Cortex-M4
Power Domain:	D2

Counter Settings:

Prescaler (PSC - 16 bits value)	prescaler *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	period *
Internal Clock Division (CKD)	No Division

auto-reload preload	Disable
Trigger Output (TRGO) Parameters:	
Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

3.27. USART1

Mode: Asynchronous

3.27.1. Parameter Settings:

Core(s) Settings:

Context(s):	Cortex-M7 Cortex-M4
Initialized Context:	Cortex-M7
Power Domain:	D2

Basic Parameters:

Baud Rate	2000000 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.28. FATFS_M7

mode: User-defined

3.28.1. Set Defines:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Version:

FATFS version	R0.12c
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Function Parameters:

FS_READONLY (Read-only mode)	Disabled
FS_MINIMIZE (Minimization level)	Disabled
USE_STRFUNC (String functions)	Enabled with LF -> CRLF conversion
USE_FIND (Find functions)	Disabled
USE_MKFS (Make filesystem function)	Enabled
USE_FASTSEEK (Fast seek function)	Enabled
USE_EXPAND (Use f_expand function)	Disabled
USE_CHMOD (Change attributes function)	Disabled
USE_LABEL (Volume label functions)	Disabled
USE_FORWARD (Forward function)	Disabled

Locale and Namespace Parameters:

CODE_PAGE (Code page on target)	Latin 1
USE_LFN (Use Long Filename)	Enabled with dynamic working buffer on the HEAP *
MAX_LFN (Max Long Filename)	255
LFN_UNICODE (Enable Unicode)	ANSI/OEM
STRF_ENCODE (Character encoding)	UTF-8
FS_RPATH (Relative Path)	Enabled with f_getcwd *

Physical Drive Parameters:

VOLUMES (Logical drives)	1
MAX_SS (Maximum Sector Size)	4096 *
MIN_SS (Minimum Sector Size)	4096 *
MULTI_PARTITION (Volume partitions feature)	Disabled
USE_TRIM (Erase feature)	Disabled
FS_NOFSINFO (Force full FAT scan)	0

System Parameters:

FS_TINY (Tiny mode)	Disabled
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FS_EXFAT (Support of exFAT file system)	Disabled
FS_NORTC (Timestamp feature)	Fixed timestamp *
NORTC_YEAR (Year for timestamp)	2015
NORTC_MON (Month for timestamp)	6
NORTC_MDAY (Day for timestamp)	4
FS_REENTRANT (Re-Entrancy)	Enabled
FS_TIMEOUT (Timeout ticks)	1000
USE_MUTEX	Disabled
SYNC_t (O/S sync object)	osSemaphoreId
FS_LOCK (Number of files opened simultaneously)	2

3.29. FREERTOS_M7

Interface: CMSIS_V1

3.29.1. Config parameters:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

API:

FreeRTOS API	CMSIS v1
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Versions:

FreeRTOS version	10.3.1
CMSIS-RTOS version	1.02

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Enabled *

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	7
MINIMAL_STACK_SIZE	256 *
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Disabled

USE_COUNTING_SEMAPHORES	Enabled *
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Enabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Enabled *
Memory management settings:	
Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	200000 *
Memory Management scheme	heap_4
Hook function related definitions:	
USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Option2 *
Run time and task stats gathering related definitions:	
GENERATE_RUN_TIME_STATS	Enabled *
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Enabled *
Co-routine related definitions:	
USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2
Software timer definitions:	
USE_TIMERS	Enabled *
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	512
Interrupt nesting behaviour configuration:	
LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5
Added with 10.2.1 support:	
MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

3.29.2. Include parameters:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled *
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Disabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Enabled *
uxTaskGetStackHighWaterMark	Enabled *
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Disabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Disabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.29.3. Advanced settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Enabled *
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Project settings (see parameter description first):

Use FW pack heap file	Enabled
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3.30. LWIP

mode: Enabled

Advanced parameters are not listed except if modified by user.

3.30.1. General Settings:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

LwIP Version:

LwIP Version (Version of LwIP supported by CubeMX ** CubeMX specific **)	2.1.2
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IPv4 - DHCP Options:

LWIP_DHCP (DHCP Module)	Disabled *
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IP Address Settings:

IP_ADDRESS (IP Address)	192.168.000.010 *
NETMASK_ADDRESS (Netmask Address)	255.255.255.000 *
GATEWAY_ADDRESS (Gateway Address)	000.000.000.000

RTOS Dependency:

WITH_RTOS (Use FREERTOS ** CubeMX specific **)	Enabled
CMSIS_VERSION (CMSIS API Version used)	CMSIS v1
RTOS_USE_NEWLIB_REENTRANT (RTOS used - 1)	Enabled

Platform Settings:

PHY Driver	Choose/LAN8742
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Protocols Options:

LWIP_ICMP (ICMP Module Activation)	Enabled
LWIP_IGMP (IGMP Module)	Enabled *
LWIP_DNS (DNS Module)	Disabled
LWIP_UDP (UDP Module)	Enabled
MEMP_NUM_UDP_PCB (Number of UDP Connections)	3 *
LWIP_TCP (TCP Module)	Enabled
MEMP_NUM_TCP_PCB (Number of TCP Connections)	15 *

3.30.2. Key Options:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1
Infrastructure - OS Awareness Option:	
NO_SYS (OS Awareness)	OS Used
Infrastructure - Timers Options:	
LWIP_TIMERS (Use Support For sys_timeout)	Enabled
Infrastructure - Core Locking and MPU Options:	
SYS_LIGHTWEIGHT_PROT (Memory Functions Protection)	Enabled
Infrastructure - Heap and Memory Pools Options:	
MEM_SIZE (Heap Memory Size)	131048 *
MEMP_SANITY_CHECK (Memory Pool Sanity Check)	Enabled *
LWIP_RAM_HEAP_POINTER (RAM Heap Pointer)	0x30020000 *
ETH_RX_BUFFER_CNT	12
Infrastructure - Internal Memory Pool Sizes:	
MEMP_NUM_PBUF (Number of Memory Pool struct Pbufs)	16
MEMP_NUM_RAW_PCB (Number of Raw Protocol Control Blocks)	4
MEMP_NUM_TCP_PCB_LISTEN (Number of Listening TCP Connections)	8
MEMP_NUM_TCP_SEG (Number of TCP Segments simultaneously queued)	16
MEMP_NUM_NETCONN (Number of Netconns Structures)	8 *
MEMP_NUM_LOCALHOSTLIST (Number of Host Entries in the Local Host List)	1
Pbuf Options:	
PBUF_POOL_SIZE (Number of Buffers in the Pbuf Pool)	16
PBUF_POOL_BUFSIZE (Size of each pbuf in the pbuf pool)	592
IPv4 - ARP Options:	
LWIP_ARP (ARP Functionality)	Enabled
Callback - TCP Options:	
TCP_TTL (Number of Time-To-Live Used by TCP Packets)	255
TCP_WND (TCP Receive Window Maximum Size)	5840
TCP_QUEUE_OOSEQ (Allow Out-Of-Order Incoming Packets)	Enabled
LWIP_TCP_SACK_OUT (Allow Sending Selective Acknowledgements)	Disabled
TCP_MSS (Maximum Segment Size)	1460 *
TCP_SND_BUF (TCP Sender Buffer Space)	5840 *
TCP_SND_QUEUELEN (Number of Packet Buffers Allowed for TCP Sender)	16 *
Network Interfaces Options:	
LWIP_NETIF_STATUS_CALLBACK (Callback Function on Interface Status Changes)	Disabled
LWIP_NETIF_EXT_STATUS_CALLBACK (Extended Callback Function for several netif)	Disabled
LWIP_NETIF_LINK_CALLBACK (Callback Function on Interface Link Changes)	Enabled

NETIF - Loopback Interface Options:

LWIP_NETIF_LOOPBACK (NETIF Loopback) Disabled

Infrastructure - Threading Options:

TCPIP_THREAD_NAME (TCPIP Thread Name) "tcpip_thread"

TCPIP_THREAD_STACKSIZE (TCPIP Thread Stack Size) **2048 ***

TCPIP_THREAD_PRIO (TCPIP Thread Priority Level) 3

TCPIP_MBOX_SIZE (TCPIP Mailbox Size) 6

DEFAULT_THREAD_NAME (Default LwIP Thread Name) "lwIP"

DEFAULT_THREAD_STACKSIZE (Default LwIP Thread Stack Size) 1024

DEFAULT_THREAD_PRIO (Default LwIP Thread Priority Level) 3

DEFAULT_RAW_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN Raw) 0

DEFAULT_TCP_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN TCP) 6

DEFAULT_ACCEPTMBOX_SIZE (Default Mailbox Size for Incoming Connections) 6

Thread Safe APIs - Netconn Options:

LWIP_NETCONN (NETCONN API) Enabled

Thread Safe APIs - Socket Options:

LWIP_SOCKET (Socket API) Enabled

LWIP_COMPAT_SOCKETS (BSD-style Socket Functions Names) 1

LWIP_SOCKET_OFFSET (Socket Offset Number) 0

LWIP_SO_SNDTIMEO (Send Timeout for Socket/Netconns) **Enabled ***

LWIP_SOCKET_SELECT (Select for Socket) Enabled

LWIP_SOCKET_POLL (Poll for Socket) Enabled

3.30.3. PPP:

Core(s) Settings:

Context(s): Cortex-M7

Initialized Context: Cortex-M7

Power Domain: D1

PPP Options:

PPP_SUPPORT (PPP Module) Disabled

3.30.4. IPv6:

Core(s) Settings:

Context(s): Cortex-M7

Initialized Context: Cortex-M7

Power Domain: D1

IPv6 Options:

LWIP_IPV6 (IPv6 Protocol) Disabled

3.30.5. HTTPD:

Core(s) Settings:

Context(s): Cortex-M7

Initialized Context: Cortex-M7

Power Domain: D1

HTTPD Options:

LWIP_HTTPD (LwIP HTTPD Support ** CubeMX specific **) Enabled *

3.30.6. SNMP:

Core(s) Settings:

Context(s): Cortex-M7

Initialized Context: Cortex-M7

Power Domain: D1

SNMP Options:

LWIP_SNMP (LwIP SNMP Agent) Disabled

3.30.7. SNTP/SMTP:

Core(s) Settings:

Context(s): Cortex-M7

Initialized Context: Cortex-M7

Power Domain: D1

SNTP Options:

LWIP_SNTP (LWIP SNTP Support ** CubeMX specific **) Disabled

SMTP Options:

LWIP_SMTP (LWIP SMTP Support ** CubeMX specific **) Disabled

3.30.8. MDNS/TFTP:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1
MDNS Options:	
LWIP_MDNS (Multicast DNS Support ** CubeMX specific **)	Enabled *
TFTP Options:	
LWIP_TFTP (TFTP Support ** CubeMX specific **)	Disabled

3.30.9. Perf/Checks:

Core(s) Settings:	
Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1
Sanity Checks:	
LWIP_DISABLE_TCP_SANITY_CHECKS (TCP Sanity Checks)	Disabled
LWIP_DISABLE_MEMP_SANITY_CHECKS (MEMP Sanity Checks)	Disabled
Performance Options:	
LWIP_PERF (Performace Testing for LWIP)	Disabled

3.30.10. Statistics:

Core(s) Settings:	
Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1
Debug - Statistics Options:	
LWIP_STATS (Statictics Collection)	Disabled

3.30.11. Checksum:

Core(s) Settings:	
Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

Infrastructure - Checksum Options:

CHECKSUM_BY_HARDWARE (Hardware Checksum ** CubeMX specific **)	Enabled
LWIP_CHECKSUM_CTRL_PER_NETIF (Generate/Check Checksum per Netif)	Disabled
CHECKSUM_GEN_IP (Generate Software Checksum for Outgoing IP Packets)	Disabled
CHECKSUM_GEN_UDP (Generate Software Checksum for Outgoing UDP Packets)	Disabled
CHECKSUM_GEN_TCP (Generate Software Checksum for Outgoing TCP Packets)	Disabled
CHECKSUM_GEN_ICMP (Generate Software Checksum for Outgoing ICMP Packets)	Enabled
CHECKSUM_GEN_ICMP6 (Generate Software Checksum for Outgoing ICMP6 Packets)	Disabled
CHECKSUM_CHECK_IP (Generate Software Checksum for Incoming IP Packets)	Disabled
CHECKSUM_CHECK_UDP (Generate Software Checksum for Incoming UDP Packets)	Disabled
CHECKSUM_CHECK_TCP (Generate Software Checksum for Incoming TCP Packets)	Disabled
CHECKSUM_CHECK_ICMP (Generate Software Checksum for Incoming ICMP Packets)	Enabled
CHECKSUM_CHECK_ICMP6 (Generate Software Checksum for Incoming ICMP6 Packets)	Disabled

3.30.12. Debug:

Core(s) Settings:

Context(s):	Cortex-M7
Initialized Context:	Cortex-M7
Power Domain:	D1

LwIP Main Debugging Options:

LWIP_DBG_MIN_LEVEL (Minimum Level)	All
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3.30.13. Platform Settings:

Driver_PHY	LAN8742
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* User modified value

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
DCMI	PI6	DCMI_D6	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PI5	DCMI_VSYN C	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PI4	DCMI_D5	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PG10	DCMI_D2	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PI7	DCMI_D7	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PC11	DCMI_D4	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PA10	DCMI_D1	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PC9	DCMI_D3	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PA6	DCMI_PIXCL K	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PH9	DCMI_D0	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PA4	DCMI_HSYN C	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
DEBUG	PA14 (JTCK/S WCLK)	DEBUG_JTC K-SWCLK	n/a	n/a	n/a		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PA13 (JTMS/S WDIO)	DEBUG_JTM S-SWDIO	n/a	n/a	n/a		Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
DSIHOST	DSI_D1P	DSIHOST_D 1P	n/a	n/a	n/a		Cortex-M4	D1
	DSI_D1N	DSIHOST_D 1N	n/a	n/a	n/a		Cortex-M4	D1
	DSI_CKP	DSIHOST_C KP	n/a	n/a	n/a		Cortex-M4	D1
	DSI_CKN	DSIHOST_C KN	n/a	n/a	n/a		Cortex-M4	D1
	DSI_D0P	DSIHOST_D 0P	n/a	n/a	n/a		Cortex-M4	D1
	DSI_D0N	DSIHOST_D 0N	n/a	n/a	n/a		Cortex-M4	D1

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
ETH	PG11	ETH_TX_EN	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PG14	ETH_TXD1	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PG13	ETH_TXD0	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PC1	ETH_MDC	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PA2	ETH_MDIO	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PA1	ETH_REF_CLK	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PA7	ETH_CRSDV	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PC4	ETH_RXD0	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
	PC5	ETH_RXD1	Alternate Function Push Pull	No pull-up and no pull-down	High		Cortex-M7	D2
FMC	PE1	FMC_NBL1	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE0	FMC_NBL0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PG15	FMC_SDNCAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PD0	FMC_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PD1	FMC_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PG8	FMC_SDCLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF2	FMC_A2	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF1	FMC_A1	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF0	FMC_A0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PG5	FMC_BA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF3	FMC_A3	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PG4	FMC_BA0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PG2	FMC_A12	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF5	FMC_A5	Alternate Function	No pull-up and no pull-	Very High		Cortex-M7*	D1

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
			Push Pull	down			Cortex-M4	
	PF4	FMC_A4	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PC0	FMC_SDNWE	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PC2	FMC_SDNE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PC3	FMC_SDCKE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE10	FMC_D7	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF13	FMC_A7	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF14	FMC_A8	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE9	FMC_D6	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE11	FMC_D8	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PD15	FMC_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PD14	FMC_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF12	FMC_A6	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF15	FMC_A9	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE12	FMC_D9	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE15	FMC_D12	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PF11	FMC_SDNRAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PG0	FMC_A10	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE8	FMC_D5	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE13	FMC_D10	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PD10	FMC_D15	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PD9	FMC_D14	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PG1	FMC_A11	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
	PE7	FMC_D4	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PE14	FMC_D11	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
	PD8	FMC_D13	Alternate Function Push Pull	No pull-up and no pull-down	Very High		Cortex-M7* Cortex-M4	D1
I2C2	PH4	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M7	D2
	PH5	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M7	D2
I2C3	PA8	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
	PH8	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low		Cortex-M4	D2
QUADSPI	PF6	QUADSPI_B K1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D1
	PF7	QUADSPI_B K1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D1
	PF8	QUADSPI_B K1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D1
	PF10	QUADSPI_C LK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D1
	PF9	QUADSPI_B K1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D1
	PB10	QUADSPI_B K1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D1
RCC	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a		Cortex-M7* Cortex-M4	D3
SPI1	PB5	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PG9	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PB3 (JTDO/TRACESWO)	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
SPI2	PD3	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D2
	PB9	SPI2_NSS	Alternate Function	No pull-up and no pull-	Very	MEMS_CS	Cortex-M7	D2

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
			Push Pull	down	High *			
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D2
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *		Cortex-M7	D2
SPI4	PE2	SPI4_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PE5	SPI4_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PE4	SPI4_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
	PE6	SPI4_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7	D2
TIM1	PA9	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	CIS_CP	Cortex-M7	D2
TIM3	PC6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	CIS_LED_B	Cortex-M7	D2
TIM4	PD13	TIM4_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	CIS_LED_R	Cortex-M7	D2
TIM5	PH12	TIM5_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	CIS_LED_G	Cortex-M7	D2
TIM8	PI2	TIM8_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	XLSYNC	Cortex-M7	D2
	PC8	TIM8_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	CIS_SP	Cortex-M7	D2
USART1	PB6	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	D2
	PB7	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low		Cortex-M7* Cortex-M4	D2
GPIO	PK5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ETH_RST	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PA15 (JTDI)	GPIO_Input	Input mode	Pull-up *	n/a	MEMS_FSYNC	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PA12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	CIS_RS	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PG6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EN_12V	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PK2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EN_5V	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4
	PJ8	GPIO_EXTI8	External Interrupt Mode with	Pull-up *	n/a	MEMS_INT	Cortex-M7* Cortex-M4	Cortex-M7* Cortex-M4

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context	Power Domain
			Falling edge					

* Initialized context

4.2. DMA configuration

DMA request	Stream	Direction	Priority
SPI2_RX	DMA2_Stream7	Peripheral To Memory	High *
SPI2_TX	DMA2_Stream6	Memory To Peripheral	High *

SPI2_RX: DMA2_Stream7 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

SPI2_TX: DMA2_Stream6 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

MDMA Request	Channel	Trigger Mode	Priority
SW	MDMA_Channel0	Buffer transfer	Low

SW: MDMA_Channel0 DMA request Settings:

Buffer Transfer Length: **2540 ***
 Data Mask: 0
 Source Burst Size: **Burst of 4 beats ***
 Destination Burst Size:

Burst of 4 beats *

Block Data Length: **2540 ***
Destination Address: 0
Destination Block Address 0
Offset:
Address Mask: 0
Data Alignment: N/A
Source Block Address 0
Offset:
Endianness: Little endianness preserved
Source Address: 0
Source Increment mode: Increment by a word
Block Count: **1 ***
Destination Increment mode: Increment by a word
Source Data Width: **Word ***
Destination Data Width: **Word ***

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
TIM2 global interrupt	true	15	0
SPI2 global interrupt	true	6	0
TIM8 trigger and commutation interrupts and TIM14 global interrupt	true	5	0
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	true	5	0
Ethernet global interrupt	true	6	0
DMA2 stream6 global interrupt	true	7	0
DMA2 stream7 global interrupt	true	7	0
QUADSPI global interrupt	true	8	0
MDMA global interrupt	true	5	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
EXTI line[9:5] interrupts	unused		
TIM1 break interrupt	unused		
TIM1 update interrupt	unused		
TIM1 trigger and commutation interrupts	unused		
TIM1 capture compare interrupt	unused		
TIM3 global interrupt	unused		
TIM4 global interrupt	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
SPI1 global interrupt	unused		
USART1 global interrupt	unused		
TIM8 break interrupt and TIM12 global interrupt	unused		
TIM8 update interrupt and TIM13 global interrupt	unused		
TIM8 capture compare interrupt	unused		
FMC global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
TIM5 global interrupt		unused	
Ethernet wake-up interrupt through EXTI line 86		unused	
CM4 send event interrupt for CM7		unused	
DCMI global interrupt		unused	
HASH and RNG global interrupts		unused	
FPU global interrupt		unused	
SPI4 global interrupt		unused	
DMA2D global interrupt		unused	
HSEM1 global interrupt		unused	
Hold core interrupt		unused	

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
TIM2 global interrupt	false	true	true
SPI2 global interrupt	false	true	true
TIM8 trigger and commutation interrupts and TIM14 global interrupt	false	true	true
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	false	true	true
Ethernet global interrupt	false	true	true
DMA2 stream6 global interrupt	false	true	true
DMA2 stream7 global interrupt	false	true	true
QUADSPI global interrupt	false	true	true
MDMA global interrupt	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0

Interrupt Table	Enable	Preenmption Priority	SubPriority
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
TIM8 break interrupt and TIM12 global interrupt	true	2	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
CM7 send event interrupt for CM4	unused		
I2C3 event interrupt	unused		
I2C3 error interrupt	unused		
HASH and RNG global interrupts	unused		
FPU global interrupt	unused		
DSI global Interrupt	unused		
HSEM2 global interrupt	unused		
Hold core interrupt	unused		

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
TIM8 break interrupt and TIM12 global interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Category view Context Execution view Context Initialization view Power Domain view

Choose filters ...

... by Context Execution: ☐ Cortex-M7 ☐ Cortex-M4

... by Context Initialization: ☐ Cortex-M7 ☐ Cortex-M4 ☒ None

... by Power Domain: ☐ D1 ☐ D2 ☐ D3 ☒ None

Middleware

FATFS_M7 ✓ FREERTOS_M7 ✓ LWIP ✓

System Core

Analog

Timers

Connectivity

Multimedia

Security

Computing

Trace and Debug

Power and Thermal

Utilities

Other

BDMA

CORTEX_M4

TIM1 ✓

ETH ✓

DCMI ✓

RNG ✓

CRC ✓

DEBUG ✓

CORTEX_M7 ✓

DMA ✓

TIM3 ✓

FMC ✓

DMA2D ✓

GPIO ✓

IWDG1 ✓

TIM4 ✓

I2C2 ✓

DSIHOST ✓

MDMA ✓

NVIC1 ✓

TIM5 ✓

I2C3 ✓

NVIC2 ✓

RCC ✓

TIM6 ✓

QUADSPI ✓

SYS ✓

SYS_M4 ✓

TIM8 ✓

SPI1 ✓

TIM12 ✓

SPI2 ✓

SPI4 ✓

USART1 ✓

5.1.2. Without filters

Category view Context Execution view Context Initialization view Power Domain view

Choose filters ...

... by Context Execution: ☐ Cortex-M7 ☐ Cortex-M4

... by Context Initialization: ☐ Cortex-M7 ☐ Cortex-M4 ☒ None

... by Power Domain: ☐ D1 ☐ D2 ☐ D3 ☒ None

Middleware

FATFS_M7 ✓

FREERTOS_M7 ✓

LWIP ✓

System Core

Analog

Timers

Connectivity

Multimedia

Security

Computing

Trace and Debug

Power and Thermal

Utilities

Other

BDMA

CORTEX_M4

TIM1 ✓

ETH ✓

DCMI ✓

RNG ✓

CRC ✓

DEBUG ✓

CORTEX_M7 ✓

DMA ✓

TIM3 ✓

FMC ✓

DMA2D ✓

GPIO ✓

IWDG1 ✓

TIM4 ✓

I2C2 ✓

DSIHOST ✓

MDMA ✓

NVIC1 ✓

TIM5 ✓

I2C3 ✓

NVIC2 ✓

RCC ✓

TIM6 ✓

QUADSPI ✓

SYS ✓

SYS_M4 ✓

TIM8 ✓

SPI1 ✓

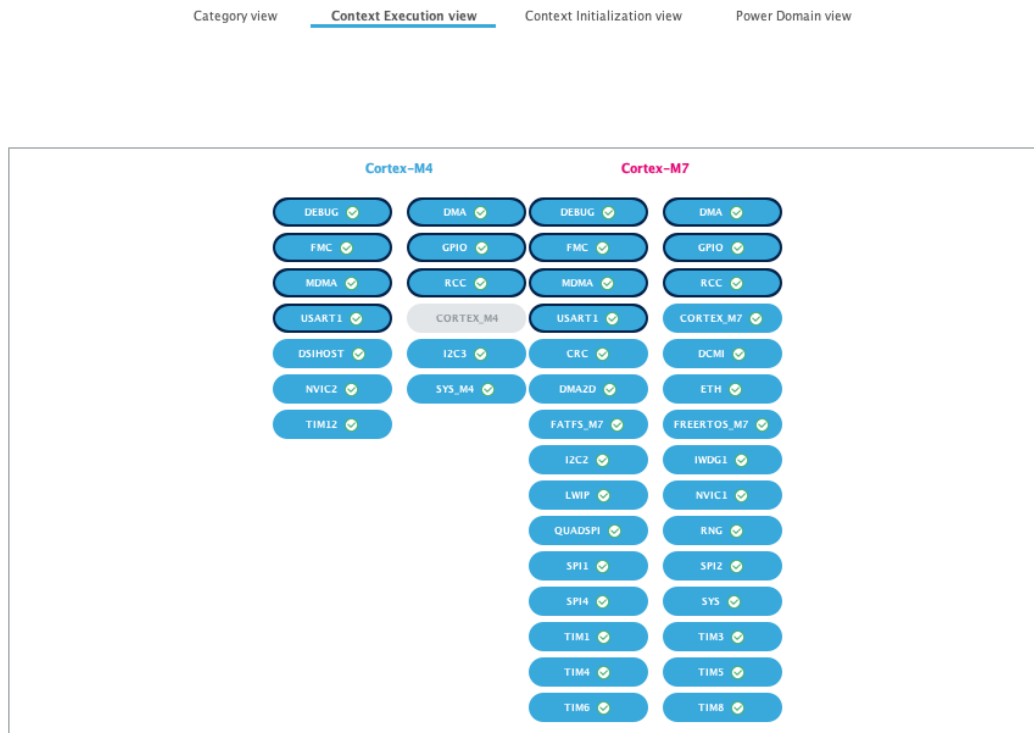
TIM12 ✓

SPI2 ✓

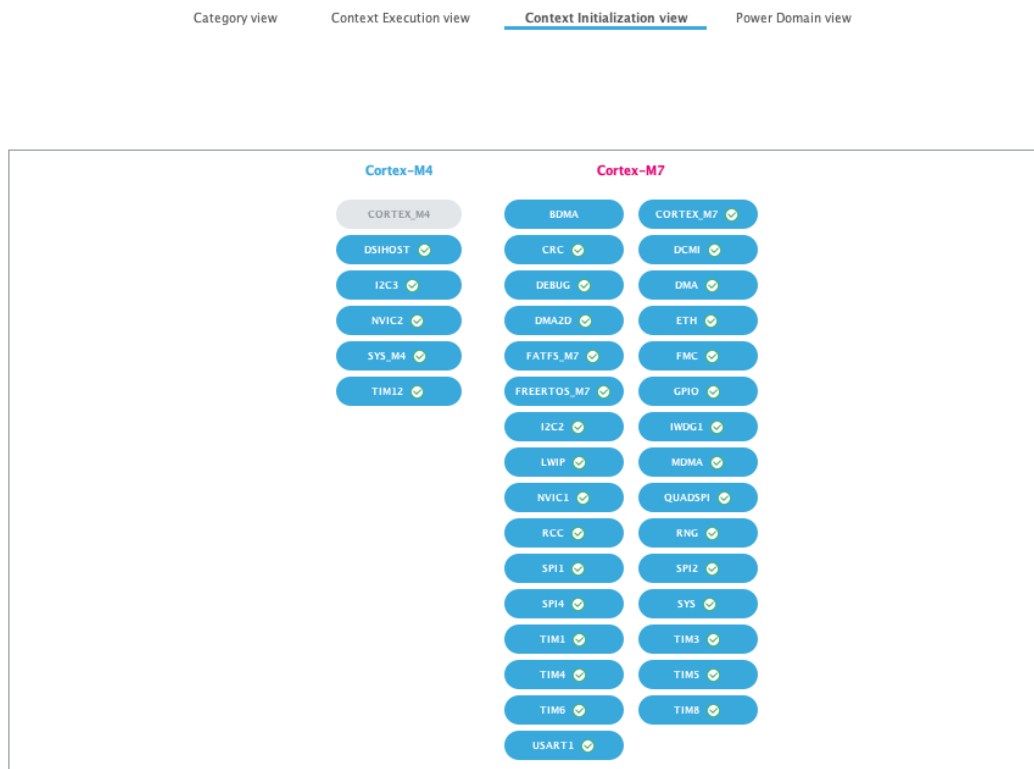
SPI4 ✓

USART1 ✓

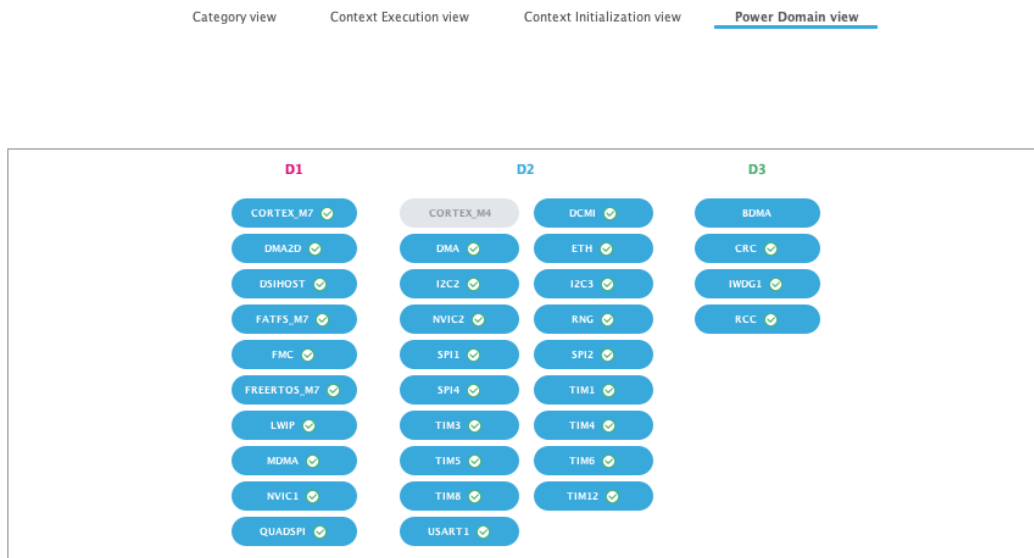
5.2. Context Execution view



5.3. Context Initialization view



5.4. Power Domain view



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdI.zip
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